

The Causal Memory and Credit Protocol

Exactly-Once Assimilation of Causal Innovation through Persistent Shared Memory, Sparse Credit Transport, and Information Geometry

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Abstract. Adaptive systems must reuse stable knowledge across contexts while continuing to learn from new evidence. Existing approaches typically conflate three operations: reading a shared mechanism, assigning credit to it, and treating an incoming signal as new evidence about it. This conflation produces two complementary failures. Local learners may not deliver useful credit to distant mechanisms, while global credit channels may rewrite many modules with correlated copies of the same causal contribution. The result is credit starvation on one side and diffuse interference, evidence double counting, and order-dependent forgetting on the other.

This paper proposes the **Causal Memory and Credit Protocol (CMCP)**, a domain-general protocol for learning systems in which shared mechanisms are represented once, reused broadly, and modified only by causally attributable innovation. CMCP separates a representation plane, a credit plane, and an evidence-ledger plane. Persistent shared memory is read-many but write-gated. A sparse credit-only highway transports mechanism-indexed learning signals without becoming an additional dependency in the forward model. An evidence ledger residualizes incoming signals against what has already been assimilated and distinguishes shared innovation, local exceptions, redundant manifestations, genuinely new mechanisms, and mechanism drift.

The protocol’s core mechanism-matching problem is formulated information-geometrically. Local mechanisms are represented by natural-gradient credit vectors and tuples of Fisher, generalized Gauss-Newton, Hessian, or linearized-dynamics operators. Metric-aware intertwiners align these operators across contexts; trace-word fingerprints provide coordinate-invariant retrieval keys; commutators and holonomy diagnose path-dependent re-encoding. A central refinement is that directional projection alone is not a valid novelty test: independent repeated evidence may lie in an already known direction and should still increase precision. CMCP therefore combines orthogonal directional novelty with a conditional-information residual, expressed locally by a Schur complement of score covariances. The resulting protocol aims for exactly-once assimilation of each causal innovation, not exactly-once use of each sample.

We state operational contracts, give a generic algorithm, describe scalable approximations based on Hessian-vector products and stochastic trace estimation, and propose falsifiable experiments spanning continual learning, reinforcement learning, world models, control, federated systems, scientific model revision, and autonomous agents.

Keywords: causal memory; credit assignment; continual learning; information geometry; Fisher information; curvature; holonomy; modular learning; evidence accounting; mechanism reuse

1. Introduction

A capable adaptive system should be able to learn a stable mechanism once, reuse it in many situations, and revise it when genuinely new evidence arrives. These goals sound compatible, but ordinary learning architectures often bind them together in a damaging way. A parameter, expert, skill, model component, or memory item that is broadly reusable is also exposed to broad plasticity. Every context that reads it may generate another update to it. Conversely, a mechanism protected from interference may become difficult to reach with useful learning signals.

This tension appears in many forms. In deep local learning, credit from a high-level discrepancy may decay before it reaches an early mechanism. In continual learning, a global gradient can alter parameters unrelated to the current causal error. In modular systems, multiple experts may learn redundant versions of the same cause. In replay systems, correlated manifestations of one event may be treated as independent evidence. In agents with long-term memory, semantically similar episodes may be repeatedly consolidated even when they contain no new information. In scientific model revision, the same anomaly may enter through several data products and be counted multiple times. These are not merely optimization failures. They are failures to distinguish **computational reuse** from **statistical assimilation**.

The motivating principle of this paper is therefore:

Shared knowledge should be broadly readable but narrowly writable; error should be globally observable but locally actionable.

The Causal Memory and Credit Protocol (CMCP) combines two architectural ideas. The first is **persistent shared causal memory**: a stable mechanism is represented once, can be queried by many contexts, and has explicit write permissions and provenance. The second is a **causal credit-only highway**: high-level discrepancy is factorized into mechanism-indexed credit, transported directly to responsible modules, and used to modify plasticity without necessarily creating a new forward or generative dependency. The highway addresses credit reach. The memory supplies the reference needed to decide what is already known. Their combination turns learning into an evidence-assimilation protocol rather than an unstructured sequence of parameter updates.

The phrase *causal* is used here in an operational, intervention-oriented sense. A module is causally responsible for a discrepancy to the extent that controlled change to that module can change the relevant outcome. CMCP does not claim that second-order geometry alone solves unrestricted causal discovery from observational data. Instead, it divides the problem into two parts:

1. **Geometry identifies and aligns candidate mechanism subspaces.** It determines whether local learning systems are the same up to a change of coordinates, whether their evidence directions overlap, and whether their transports are path-dependent.
2. **Intervention or credible invariance evidence orients causal responsibility.** It determines whether changing a candidate module can actually reduce the discrepancy and therefore whether the module may receive write credit.

This division makes the difficult matching problem substantially more tractable. Around an operating point, a mechanism is not represented only by an activation embedding. It is represented by its local sensitivity, curvature, contraction, and interference structure. Natural-gradient Hessians, generalized Gauss-Newton matrices, empirical Fisher operators, Jacobian Gramians, and linearized transition operators are all possible second-order signatures. Under suitable regularity, tuples of self-adjoint operators can be matched up to isometry by coordinate-invariant trace data; their commutators expose the failure of simultaneous alignment and the curvature of the task connection [1], [2].

CMCP also sharpens what *novel evidence* must mean. A naive memory system might project incoming credit onto the complement of the stored mechanism subspace and discard the parallel component. That is wrong. A second independent measurement of the same mechanism lies in the same direction but should increase confidence. An exact duplicate or a correlated descendant of a previously assimilated observation should not. The correct distinction is not simply *new direction versus old direction*. It is:

- **directional novelty:** does the evidence concern a previously unrepresented mechanism direction?
- **evidential novelty:** conditional on the information already stored, does the observation carry additional information within an existing mechanism direction?

We give a local information-geometric answer using residualized score variables and a conditional Fisher information matrix. Exact duplicate manifestations have zero conditional information; conditionally independent repeated observations retain their information; partial redundancy is attenuated continuously.

The paper makes six contributions.

1. It defines CMCP as a three-plane architecture separating representation, credit transport, and evidence accounting.
2. It formulates mechanism identity and transport through information geometry, operator intertwiners, invariant fingerprints, commutators, and holonomy.
3. It introduces a two-part novelty operator combining orthogonal mechanism discovery with conditional information within known mechanisms.
4. It states protocol contracts for causal idempotence, independent-evidence accumulation, modular commutation, coordinate equivariance, path consistency, and write safety.
5. It gives a generic algorithm and practical approximations that do not require dense Hessians.
6. It proposes a cross-domain experimental program designed to falsify the protocol’s core claims rather than merely report end-task accuracy.

The proposal builds on a broader geometric view of continual learning in which metric anisotropy and connection curvature are distinct failure channels [3], [4], and on the idea of a causally designed error highway that transmits novelty-filtered residuals to modules with evidence of causal responsibility [5]. The present paper elevates these ideas into a general learning protocol and adds the information-geometric mechanism-matching and conditional-novelty machinery needed to make the combination coherent.

2. Problem Setting

2.1 Adaptive systems, contexts, and mechanisms

Let an adaptive system encounter a sequence of contexts $c \in \mathcal{C}$. A context may be a task, environment, user, scientific regime, sensory condition, episode, dataset shard, control mode, or conversational situation. The system contains a collection of candidate mechanisms or modules $m \in \mathcal{M}$. A module may be a parameter block, expert, skill, transition model, rule, program fragment, memory record, controller component, or causal hypothesis.

The system produces an outcome through a context-dependent computation

$$y = f_c(x; \theta_S, \theta_{L,c}),$$

where θ_S denotes parameters of persistent shared mechanisms and $\theta_{L,c}$ denotes context-local or provisional parameters. A discrepancy signal e_c may be a supervised error, temporal-difference error, prediction er-

ror, reconstruction residual, constraint violation, reward-model critique, model-checking counterexample, or mismatch between experimental prediction and observation.

A conventional learner maps e_c to a parameter update. CMCP inserts an explicit protocol between them. The protocol must answer four questions:

1. **Identity:** Which stored mechanism, if any, does this signal concern?
2. **Responsibility:** Which modules can causally change the relevant outcome?
3. **Novelty:** What part of the signal has not already been assimilated?
4. **Destination:** Should the innovation revise shared memory, remain local, create a new mechanism, or trigger a branch for drift?

These questions are logically distinct. Treating them as one gradient operation is the source of many failure modes.

2.2 Four recurring failure modes

Credit starvation

When learning is implemented through local reciprocal interactions, a high-level discrepancy may reach a distant module only through the inverse of a local self-adjoint operator. Such inverses are localized when the operator is well conditioned; preserving long reach can require slow, nearly critical dynamics [3], [6], [7]. The system then faces a reach-versus-conditioning tradeoff: either inference converges quickly and distant mechanisms receive little task information, or information propagates far only through slow relaxation.

Diffuse credit and entanglement

A naive solution is to broadcast the full discrepancy to many modules. This improves reach, but unrelated modules receive the same plasticity signal. Shared causes can be redundantly encoded through several routes, task updates become noncommuting, and subsequent learning depends on curriculum order. Direct feedback methods demonstrate that long-range learning channels can be useful, but a metric-blind broadcast does not by itself solve causal routing [5], [8].

Evidence miscounting

Even if all updates use the same coordinates and commute, the system may assign the wrong effective multiplicity to evidence. Replay, duplicated records, correlated sensor products, multiple descendants of one event, or repeated traversal of a cached trajectory can give one innovation excessive precision. Conversely, aggressive protection can undercount legitimate new evidence. This is a symmetric accounting problem: the system uses a known direction too many or too few effective times [4].

Frame mismatch and path-dependent reuse

The same causal mechanism may be represented differently across contexts. Updating in context s can change the coordinates in which context t expresses its evidence. If the corresponding local update operators do not commute, the meaning of the second update depends on the first. Infinitesimally this is curvature; over a finite context loop it is holonomy [2], [3]. This is not simply duplicate evidence. It is a failure to recognize the same mechanism after transport through different representational frames.

2.3 The design objective

CMCP aims to provide all of the following simultaneously:

- long-range credit reach without full-error broadcast;
- reusable shared mechanisms without repeated assimilation of the same innovation;
- accumulation of genuinely independent evidence even when it concerns an existing direction;
- local storage of exceptions and context-specific residuals;
- coordinate-invariant mechanism matching across representations;
- path-consistent transport when possible, and explicit curvature accounting when it is not;
- auditable write decisions with provenance and reversible consolidation.

The desired endpoint is not zero coupling. It is **sparse, explicit, and interpretable coupling**: strong communication through a small shared causal backbone, weak cross-talk elsewhere, and a ledger that records why each protected shared mechanism changed.

3. The Three-Plane Architecture

CMCP separates learning into a representation plane, a credit plane, and an evidence-ledger plane. The separation is conceptual; implementations may share data structures or hardware.

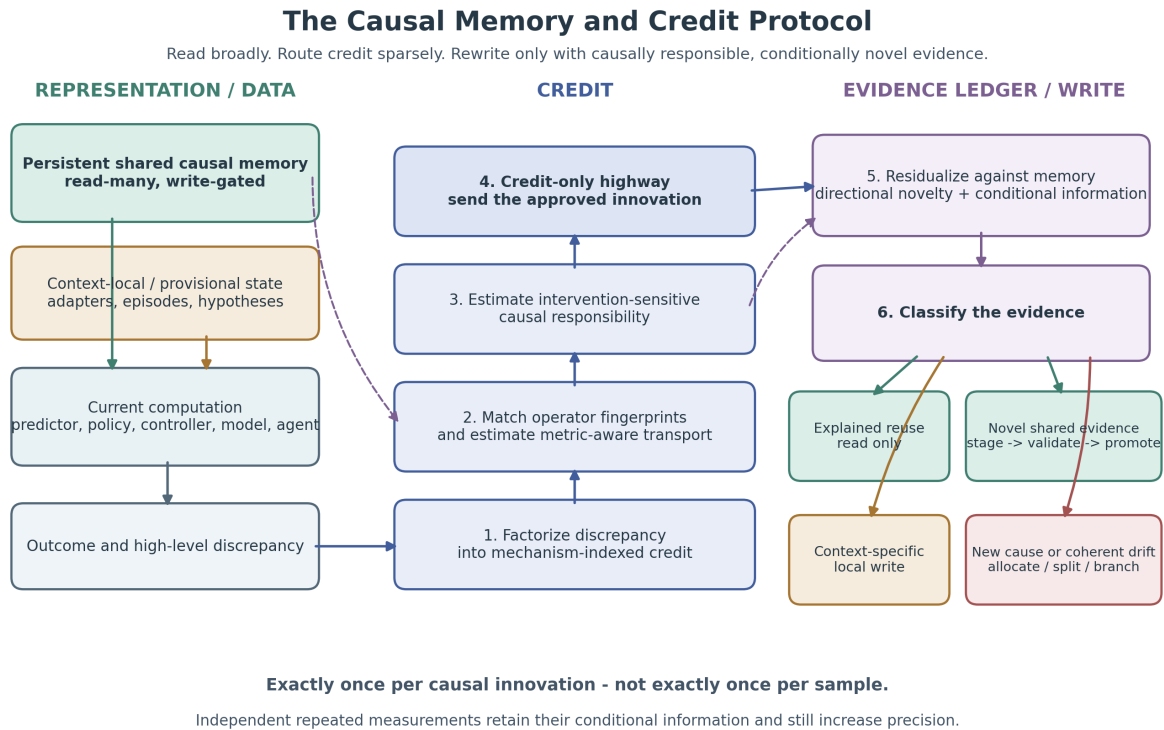


Figure 1: The three-plane CMCP architecture. Persistent memory and local modules participate in forward computation. High-level discrepancies are encoded, geometrically aligned, causally routed, novelty-filtered, and delivered through a credit-only highway. The evidence ledger decides whether to suppress, localize, provisionally store, promote, branch, or allocate.

3.1 Representation or data plane

The representation plane performs the task. It contains two broad classes of adaptive state.

Persistent shared causal memory. A shared memory item represents a stable mechanism that many contexts may query. It is read-heavy and write-gated. Its state includes not only model parameters but also an information ledger, a geometric signature, provenance, uncertainty, and lifecycle status.

Context-local or provisional state. Local adapters, episodic slots, expert-specific parameters, cached plans, or temporary hypotheses absorb context-specific evidence. They are easier to modify and discard. They protect the shared core from premature consolidation.

A useful memory record has the abstract form

$$M_j = (\theta_j, \mathcal{S}_j, \Lambda_j, \chi_j, \mathcal{P}_j, v_j),$$

where θ_j is the represented mechanism, $\mathcal{S}_j$ is its local information subspace, Λ_j is an accumulated precision or evidence ledger, χ_j is a coordinate-invariant geometric fingerprint, $\mathcal{P}_j$ is provenance, and v_j is a version or consolidation state.

The central architectural distinction is:

Reading a mechanism is a computational operation. Writing it is an evidential claim.

Many contexts may use M_j during forward computation. Only a signal that passes identity, responsibility, novelty, and safety tests may revise it.

3.2 Credit plane

The credit plane converts a high-level discrepancy into sparse, mechanism-indexed learning signals. Its main components are:

1. **Credit encoder or factorizer.** It decomposes a discrepancy into candidate components

$$z_c = C_\phi(e_c, c) = (z_{c,1}, \dots, z_{c,K}),$$

intended to separate shared, local, and nuisance contributions.

2. **Geometric matcher.** It compares local information operators with memory fingerprints, retrieves candidate mechanisms, and estimates metric-aware transport maps.
3. **Responsibility router.** It estimates whether intervention on a candidate module can reduce a component of the discrepancy.
4. **Credit-only highway.** It sends the residualized innovation directly to selected modules as a plasticity signal.

The highway is **credit-only** when it changes learning updates, eligibility traces, precision, or consolidation gates without becoming a new dependency in the forward generative model. This is the safest default because it improves credit reach while preserving the represented computation and its fixed points as much as possible [5]. A state-coupling highway may be justified when the long-range dependency is itself a real shared mechanism, but it creates a stronger burden of audit.

For a receiving module m , a generic highway update is

$$\Delta\theta_m^{\text{hw}} = -\eta b_{m,c} R_{m,c} T_{m \leftarrow c} \mathcal{N}_m(z_c),$$

where $R_{m,c}$ is a responsibility gate, $T_{m \leftarrow c}$ transports the credit into the module’s coordinates and metric, $\mathcal{N}_m$ is the novelty operator, and $b_{m,c}$ enforces bandwidth or write budgets.

3.3 Evidence-ledger or write plane

The evidence-ledger plane decides what the incoming signal means statistically. It performs the following operations:

1. retrieve candidate memory records;
2. align the incoming information geometry with each candidate;
3. estimate causal responsibility;
4. residualize against already assimilated evidence;
5. classify the residual;
6. stage, commit, merge, split, branch, or reject a write;
7. update provenance and diagnostic state.

The classification has at least five outcomes.

Incoming signal	Interpretation	Default action
High mechanism match, negligible conditional information	Repeated manifestation or already explained reuse	Read only; no shared rewrite
High mechanism match, substantial conditional information	New evidence about an existing shared mechanism	Provisional shared write; update precision
Responsible but context-specific residual	Exception, regime, policy, or local mapping	Write to local adapter or episodic slot
Low match with coherent responsibility	New mechanism	Allocate a new memory item
High prior match but persistent coherent geometric drift	Mechanism change or nonstationarity	Branch/version, then reconcile

This write plane is what allows the same highway to carry strong long-range credit without converting every global discrepancy into a global parameter update.

3.4 Two-stage commitment

A robust implementation should normally use a two-stage commit.

Stage 1: provisional assimilation. New information is placed in a local adapter, shadow copy, or provisional memory slot. The system records its geometric signature, responsibility estimate, and provenance but does not yet overwrite the protected shared mechanism.

Stage 2: promotion. The provisional innovation is promoted when it recurs across sufficiently diverse contexts, survives intervention tests, remains stable under coordinate transport, and improves prediction or control without creating broad interference.

Promotion may merge the provisional slot into shared memory, create a new shared module, or leave it local. This delay limits memory poisoning, router self-confirmation, and premature treatment of a transient correlation as a shared cause.

4. Information-Geometric Formulation

The core technical problem is to recognize a mechanism across contexts that may use different parameterizations, activations, features, or local modules. Raw parameter distance and embedding cosine similarity are not reliable because they change under reparameterization and can ignore how the system actually learns. CMCP instead compares local learning geometry.

4.1 Local credit and information operators

Let Θ be a smooth parameter manifold equipped with a positive metric g . In a statistical model, g is naturally the Fisher-Rao metric [9], [10], [11]. In a control system it may be a controllability or sensitivity metric. In a deterministic learner it may be a generalized Gauss-Newton, empirical Fisher, Jacobian Gram, or carefully chosen preconditioner.

For context c with loss or discrepancy functional L_c , define the metric credit vector

$$u_c = -\text{grad}_g L_c = -g^{-1}dL_c,$$

and the local information operator

$$H_c = \nabla^g \text{grad}_g L_c.$$

Near a local solution, H_c is the natural-gradient Hessian. It is self-adjoint in g and, for a locally fitted loss, is typically positive semidefinite. Training for a short duration h acts on a perturbation approximately as $\exp(-hH_c)$.

A single operator describes only one local quadratic form. Mechanism identity is better represented by a tuple

$$\mathbf{H}_c = (H_{c,a})_{a \in \mathcal{A}},$$

where a indexes output components, controlled probes, environments, interventions, temporal phases, or neighboring contexts. The tuple captures how a candidate mechanism responds across a small family of conditions.

4.2 Mechanism identity up to geometric gauge

Suppose contexts c and d realize the same mechanism in local spaces E_c and E_d . The natural equivalence is not equality of parameter vectors. It is the existence of an isometry or well-conditioned partial isometry $Q : E_c \rightarrow E_d$ such that

$$H_{d,a} \approx QH_{c,a}Q^*, \quad a \in \mathcal{A}.$$

This is simultaneous operator alignment. It says that the two local learning systems are the same up to their natural coordinate gauge.

For finite-dimensional tuples of self-adjoint operators, all noncommutative trace words

$$\text{Tr } w(\mathbf{H}) = \text{Tr}(H_{a_1} H_{a_2} \cdots H_{a_k})$$

form a complete invariant for simultaneous orthogonal equivalence when taken to sufficient degree. Equivalently, curriculum characters obtained from traces of products of local flow operators determine the tuple up to isometry under the conditions developed in the transfer-rigidity analysis [1]. This suggests a practical, coordinate-invariant memory key:

$$\chi_L(\mathbf{H}) = \{\text{Tr } w(\mathbf{H}) : |w| \leq L\}.$$

Complete high-degree signatures are expensive, but low-degree fingerprints are already informative in generic settings. A coarse retrieval index can include

- $\text{Tr}(H_a)$ and $\text{Tr}(H_a^2)$;
- pairwise overlaps $\text{Tr}(H_a H_b)$;
- commutator energies $\|[H_a, H_b]\|^2$;
- selected degree-three and degree-four trace words;
- low-rank spectral summaries and subspace principal angles.

The geometric key does not have to replace semantic or content-based retrieval. It can be used after a semantic shortlist to answer the more specific question: *do these candidate records implement the same local learning mechanism?*

4.3 Metric-aware transport as approximate intertwining

After retrieving a candidate memory item M_j , the protocol estimates a transport map $T_{j \leftarrow c}$. A generic objective is

$$\mathcal{L}_{\text{align}}(T) = \sum_{a \in \mathcal{A}} \|H_{j,a} T - T H_{c,a}\|_{F,g}^2 + \lambda_g \|T^* g_j T - g_c\|_F^2 + \lambda_r \Omega(T).$$

The first term asks T to intertwine the operator tuples. The second makes it approximately metric preserving. The regularizer Ω can enforce low rank, sparsity, locality, orthogonality, or a partial-isometry constraint. When dimensions differ, T maps only the matched mechanism subspace.

This transport is the connection of the protocol. It allows an innovation expressed in context c to be compared with and applied to a memory record written in another frame. Exact intertwining is a strong condition; approximate intertwining is enough to provide a confidence score and an error budget.

4.4 Symmetric information and antisymmetric path dependence

The self-adjoint operators H_c lie in the symmetric part of the local operator algebra. Their commutators

$$C_{st} = [H_s, H_t]$$

are anti-self-adjoint. This is the Cartan split

$$\mathfrak{gl}(E) = \text{Sym}(E, g) \oplus \mathfrak{so}(E, g).$$

The split has a direct protocol interpretation.

- The **symmetric component** records contraction, precision, evidence accumulation, and the strength of local learning directions. It belongs in the memory ledger.
- The **antisymmetric component** records rotation, order dependence, and the obstruction to using one common coordinate frame. It belongs in the transport and curvature audit.

For a curriculum $\Gamma = (c_1, h_1) \cdots (c_N, h_N)$, the linearized update is

$$M(\Gamma) = e^{-h_N H_{c_N}} \cdots e^{-h_1 H_{c_1}}.$$

Its polar decomposition

$$M(\Gamma) = \text{Hol}(\Gamma) P(\Gamma)$$

separates a positive symmetric factor P , describing convergence and accumulated contraction, from an orthogonal factor Hol , describing order-dependent rotation [2]. CMCP treats these as separate control channels. Precision and schedule methods correct the symmetric ledger. Causal modularization and transport methods correct the connection.

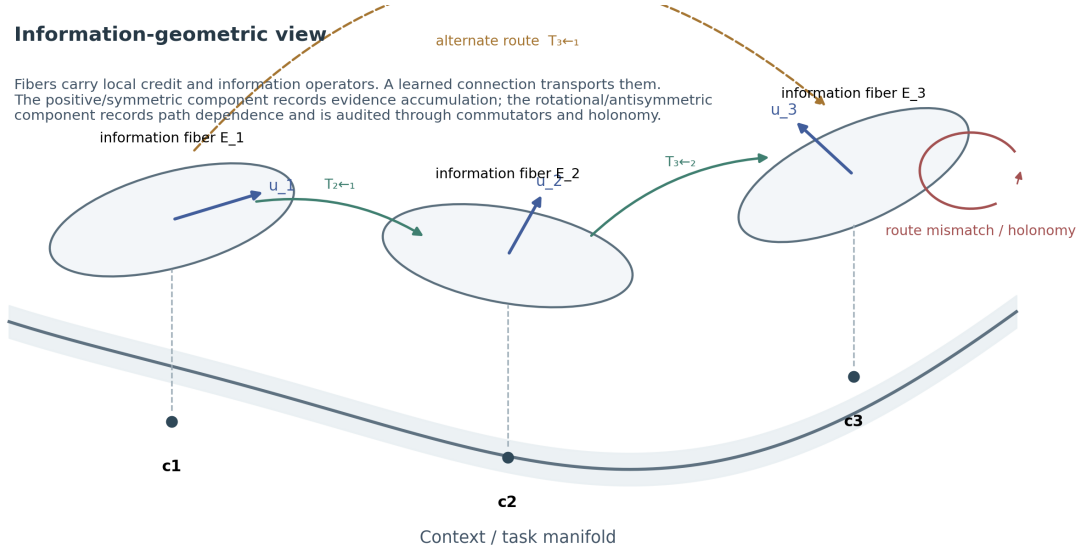


Figure 2: Information-geometric interpretation of CMCP. Contexts form a base manifold; local credit and information operators live in fibers; a learned connection transports them. Route mismatch is holonomy.

4.5 Curvature, commutators, and the interference Gram

For a small two-context loop, the leading rotational defect is proportional to the commutator:

$$\log \text{Hol}(s, t) \approx \frac{1}{2} h_s h_t [H_s, H_t].$$

Under a one-statistical-model, many-operating-points approximation, this commutator is the Fisher-Rao curvature contracted against the area spanned by the task displacements, up to model and higher-order defects [3], [4]. The commutator is therefore a local measure of whether two contexts re-express shared evidence in incompatible frames.

For many contexts, define the interference Gram

$$G_{(ij),(kl)} = -\text{Tr}([H_i, H_j][H_k, H_l]).$$

Its diagonal measures the strength of each pairwise conflict. Its off-diagonal entries measure whether different conflicts occupy the same rotation planes. A low-rank Gram indicates that one shared transport correction or module boundary may repair several conflicts; a near-diagonal Gram suggests separate interventions [2].

The goal is not necessarily to make every commutator zero. Genuine shared mechanisms can create structured overlap. The goal is to transform broad, incoherent interference into a small set of explicit shared planes and make unrelated mechanisms approximately commute.

5. Novelty: Direction Is Not Evidence

Mechanism matching answers *where* an incoming signal belongs. It does not yet answer whether the signal contains new evidence. This section develops the central evidence-accounting rule.

5.1 Why orthogonal projection is insufficient

Let $\mathcal{S}_j$ be the information subspace already associated with memory item M_j . After transport, a simple residual would be

$$\nu_{j,c}^\perp = (I - \Pi_{\mathcal{S}_j}^{g_j})T_{j \leftarrow c}\nu_c.$$

This correctly detects a new mechanism direction. But it incorrectly treats all evidence parallel to $\mathcal{S}_j$ as redundant. Consider two cases:

- the incoming record is an exact copy of a previously assimilated observation;
- the incoming record is an independent new measurement generated by the same mechanism.

Both lie in the same local direction. The first should not increase precision. The second should. Directional projection cannot distinguish them.

CMCP therefore separates **directional novelty** from **evidential novelty**.

5.2 Conditional score residualization

Let s_c be a local score or sensitivity statistic for the incoming evidence after transport into the candidate memory frame. Let s_M be a vector of score statistics or sufficient-statistic summaries already represented in the memory ledger. Define

$$F_{cc} = \mathbb{E}[s_c s_c^*], \quad F_{MM} = \mathbb{E}[s_M s_M^*], \quad F_{cM} = \mathbb{E}[s_c s_M^*].$$

The best linear prediction of s_c from stored evidence in the information metric is

$$\hat{s}_c = F_{cM} F_{MM}^\dagger s_M.$$

The residual score and its conditional information are

$$\tilde{s}_{c|M} = s_c - F_{cM} F_{MM}^\dagger s_M,$$

$$F_{c|M} = F_{cc} - F_{cM} F_{MM}^\dagger F_{Mc}.$$

The second expression is the Schur complement of the stored score block. It is positive semidefinite under the usual covariance assumptions. It quantifies locally how much information remains after linearly conditioning on what memory already represents.

This gives three important regimes.

Exact duplicate manifestation. If $s_c = As_M$ almost surely for a stored linear map A , then $F_{c|M} = 0$. The incoming record has no residual information and should not rewrite shared memory.

Independent repeated observation. If the new observation concerns the same mechanism but is conditionally independent of stored observations given the mechanism, then $F_{cM} = 0$ in the idealized score representation and $F_{c|M} = F_{cc}$. Its full information should be added to the precision ledger.

Partial redundancy. If the observation is correlated with stored evidence but not determined by it, then

$$0 \preceq F_{c|M} \preceq F_{cc}.$$

The protocol discounts only the predictable component.

This is an L^2 local residual. In non-Gaussian or strongly nonlinear settings it can be replaced or supplemented by conditional mutual information, kernel conditional covariance, learned density-ratio residualization, or a predictive coding model of stored evidence. The key requirement is the same: novelty must be conditioned on the evidence ledger, not inferred from direction alone.

5.3 The innovation packet

For candidate memory M_j , CMCP forms an innovation packet

$$\mathcal{J}_{j,c} = \left(\nu_{j,c}^\perp, \tilde{s}_{c|M_j}, F_{c|M_j}, r_{j,c}, q_{j,c} \right),$$

where $r_{j,c}$ is causal responsibility and $q_{j,c}$ summarizes transport and match quality.

The first component opens new mechanism directions. The second and third update precision and sufficient statistics within an existing direction. The last two control whether the write is causally justified and geometrically reliable.

A local quadratic or exponential-family memory can update natural parameters as

$$\Lambda_j^+ = \Lambda_j + \alpha r_{j,c} T_{j \leftarrow c} F_{c|M_j} T_{j \leftarrow c}^*,$$

$$\eta_j^+ = \eta_j + \alpha r_{j,c} T_{j \leftarrow c} \tilde{q}_{c|M_j},$$

with θ_j^+ recovered from the memory’s natural parameterization. In a generic optimizer, the same idea becomes a preconditioned proximal step using Λ_j as the consolidation metric.

The rule is deliberately asymmetric between reads and writes. A context may read the full shared mechanism even when $F_{c|M_j} = 0$. It simply does not add another copy of the same evidence.

5.4 Exactly once per causal innovation

The phrase **exactly-once assimilation** can be misunderstood. It does not mean that a data point may be processed only once. It means:

Each causally distinct innovation should contribute once to the shared evidence ledger, while each conditionally independent measurement may contribute its own information.

This is closer to deduplicated sufficient-statistic accumulation than to sample suppression. It preserves statistical efficiency while preventing a single innovation from gaining artificial confidence through many correlated representations.

5.5 Drift, contradiction, and branching

A persistent discrepancy with high responsibility can mean several different things:

- the current memory is uncertain and should gain precision;
- a context variable is missing;
- the environment has changed;
- two regimes require separate mechanisms;
- the observation is corrupted or adversarial.

CMCP should not force all such cases into one wider or less certain shared state. The geometric and evidential signatures help distinguish them.

- **Stable operator signature, new conditional information:** update the existing mechanism.
- **Stable shared core plus systematic local residual:** create or retain a context-local adapter.
- **Coherent temporal drift of the operator tuple:** branch the mechanism version and maintain a transition model between versions.
- **Multimodal incompatible signatures:** split or allocate separate mechanisms.
- **Low responsibility or poor provenance:** quarantine or reject the write.

This lifecycle treats memory revision as model selection under causal and geometric constraints rather than as unrestricted parameter motion.

6. Causal Responsibility and Sparse Routing

Information geometry can identify candidate mechanism equivalence classes and coordinate transforms. It does not, by itself, determine the causal arrow. A symmetric Hessian reveals interaction and sensitivity, not whether intervention on a module changes the outcome of interest. CMCP therefore requires a responsibility estimator.

6.1 Intervention-sensitive responsibility

A conceptual responsibility score is

$$r_{m,c,j} \approx \left\| \frac{\partial \text{do}(y_j)}{\partial \text{do}(h_m)} \right\|,$$

where h_m is the represented variable or output of module m . Practical estimates can use:

- controlled parameter or activation perturbations;
- counterfactual prediction by a learned world model;
- environment invariance and mechanism-change tests;
- influence functions or Jacobian-vector products;
- sparse do-influence masks learned under intervention supervision;
- temporal precedence plus action-conditioned transition evidence;
- randomized routing probes with held-out effect measurement.

The score should be calibrated against an observed reduction in the relevant discrepancy, not merely against gradient magnitude. A module can have a large correlational gradient because it is a descendant or proxy while lacking the capacity to intervene on the source of error.

6.2 Factorized credit

The high-level discrepancy should be decomposed before routing:

$$e_c = e_c^{\text{shared}} + e_c^{\text{local}} + e_c^{\text{nuisance}}.$$

This decomposition may be explicit in a structured model or learned through sparse latent factors. Shared components are candidates for persistent memory. Local components belong in adapters or episodic state. Nuisance components are suppressed or routed only to noise models.

A useful training objective combines task performance with routing supervision:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{do}} \mathcal{L}_{\text{responsibility}} + \lambda_{\text{path}} \mathcal{L}_{\text{path}} + \lambda_{\text{nov}} \mathcal{L}_{\text{novelty}} + \lambda_{\text{shadow}} \mathcal{L}_{\text{shadow}} + \lambda_{\text{budget}} \mathcal{L}_{\text{sparsity}}.$$

The path term penalizes disagreement between alternative transport routes. The novelty term penalizes rewriting memory with predictable evidence. The shadow term penalizes unintended commutators outside designated shared subspaces. The budget limits active routes and write bandwidth.

6.3 Why the highway should usually be credit-only

A long-range state connection changes the function or dynamical system being represented. That may be desirable, but it can also create a new causal shortcut and entangle mechanisms that the memory architecture was intended to separate. A credit-only highway instead modifies local plasticity, eligibility, precision, or inference updates while leaving the forward graph unchanged during ordinary operation.

This distinction generalizes beyond neural networks.

- In reinforcement learning, a highway may carry a factorized TD innovation to a dynamics module without adding that module as a new policy input.

- In a scientific model, it may route an anomaly to the responsible submodel without adding an empirical correction term to every prediction.
- In a database-backed agent, it may change consolidation priority without injecting the retrieved memory into all future prompts.
- In control, it may update a local model or gain schedule without adding a permanent sensor-to-actuator shortcut.

A state-coupling extension should be promoted only when the new dependency has passed the same causal, novelty, and invariance tests as any other shared mechanism.

7. Protocol Contracts

A protocol is useful when it defines behavior that can be tested independently of a particular implementation. CMCP proposes the following contracts.

7.1 Causal idempotence

For a repeated manifestation d whose conditional information given memory is negligible,

$$U_d(U_d(M)) \approx U_d(M).$$

The approximation allows numerical noise and bounded adaptation, but the second presentation should not add another full unit of shared precision.

7.2 Independent-evidence accumulation

For a conditionally independent observation d' of an existing mechanism,

$$\Lambda(U_{d'}(M)) \approx \Lambda(M) + F_{d'|M}.$$

CMCP is not an anti-repetition heuristic. It must retain the ability to become more certain when new measurements arrive.

7.3 Commutation of independent mechanisms

If two innovations concern causally independent modules with disjoint effective supports,

$$U_i U_j(M) \approx U_j U_i(M).$$

Order dependence is then confined to genuinely shared or misfactorized components. This contract is measurable through commutators, curriculum reversals, and holonomy.

7.4 Coordinate equivariance

Let Φ be a smooth reparameterization with pushforward Φ_* . The protocol should satisfy

$$\Phi_* U(p, M) \approx U(\Phi_* p, \Phi_* M).$$

Natural gradients, metric projections, operator intertwiners, and trace-word fingerprints are used precisely because they support this contract better than raw Euclidean parameter comparisons.

7.5 Path consistency

For two routes γ_1 and γ_2 transporting the same innovation from context c to memory m ,

$$\|T_{m \leftarrow c}^{\gamma_1} z - T_{m \leftarrow c}^{\gamma_2} z\|_{g_m} \leq \epsilon_{\text{path}}.$$

When curvature is nonzero, the residual cannot be driven to zero globally. The protocol should then expose the holonomy budget, route through a designated shared subspace, or allocate context-specific adapters.

7.6 Write safety and provenance

Every protected write must be attributable to a context, a responsibility estimate, a novelty estimate, a geometric match, and a lifecycle decision. Read permission does not imply write permission. Shared writes must be reversible or versioned until consolidated.

7.7 Bounded interference

The system should maintain a budget on unintended cross-module curvature. A practical condition is

$$\sum_{(i,j) \notin E_{\text{allowed}}} \| [H_i, H_j] \|_{F,g}^2 \leq B_{\text{shadow}}.$$

The allowed set contains explicitly shared mechanisms. The purpose is not to eliminate all overlap but to localize it.

8. A Generic CMCP Algorithm

The following algorithm is intentionally agnostic about model class.

Inputs: context c , observation or transition x , high-level discrepancy e_c , current shared memory $\mathcal{M}_S$, local memory $\mathcal{M}_L$, write budget B .

Outputs: task action or prediction; local and provisional updates; zero or more audited shared-memory commits.

1. **Run the current computation.** Read relevant shared and local modules and produce the output.
2. **Encode credit.** Factorize e_c into candidate mechanism components $z_{c,k}$.
3. **Construct local geometry.** Estimate ν_c , one or more local information operators $H_{c,a}$, and score or sufficient-statistic features s_c .
4. **Retrieve candidates.** Use semantic keys and the low-degree invariant fingerprint $\chi_L(\mathbf{H}_c)$ to select a small candidate set of memories.
5. **Align candidates.** For each candidate M_j , estimate $T_{j \leftarrow c}$ and an intertwining error $q_{j,c}$.
6. **Estimate responsibility.** Probe or predict the effect of intervening on M_j ; compute $r_{j,c}$.
7. **Residualize evidence.** Compute directional novelty $\nu_{j,c}^\perp$ and conditional information $F_{c|M_j}$.

8. **Classify the evidence.** Select read-only reuse, shared update, local update, allocation, split, or drift branch.
9. **Route credit.** Send only the approved innovation packet through the credit-only highway to selected modules.
10. **Stage writes.** Apply new shared evidence to a provisional buffer; update local adapters immediately within their own budget.
11. **Validate.** Test cross-context recurrence, intervention effects, path consistency, task benefit, and unintended commutators.
12. **Commit or reject.** Promote, merge, branch, freeze, reopen, or discard provisional state; append provenance.
13. **Audit.** Record evidence multiplicity, conditional-information gain, router precision, transport error, interference Gram, and retention outcomes.

A compact pseudocode form is:

```
CMCP-UPDATE(context c, discrepancy e, memory M):
  z          <- factorize_credit(e, c)
  (u, H, s)   <- local_geometry(c, z)
  candidates  <- retrieve(M, semantic_key(c), trace_fingerprint(H))

  packets <- []
  for memory item j in candidates:
    T, align_error <- estimate_transport(H, j.operator_tuple)
    responsibility <- intervention_score(j, c, z)
    dir_novelty    <- (I - projector(j.subspace)) T u
    score_resid, F_cond <- conditional_information(T s, j.ledger)
    packets.append(classify(j, dir_novelty, score_resid, F_cond,
                           responsibility, align_error))

  selected <- sparse_budgeted_route(packets)
  apply_credit_only_updates(selected)
  stage_provisional_writes(selected)
  validate_against_interventions_paths_and_interference()
  commit_merge_branch_or_reject()
  update_provenance_and_audit_log()
```

9. Scalable Approximation

The full formulation uses operator tuples and information metrics, but dense second-order matrices are not required.

9.1 Matrix-free operator access

Hessian-vector and generalized Gauss-Newton-vector products can be computed through automatic differentiation without constructing a Hessian [12]. In other adaptive systems, equivalent products arise from linearized simulation, finite-difference perturbations, adjoint methods, or system-identification probes.

A commutator action requires only nested products:

$$[H_i, H_j]v = H_i(H_jv) - H_j(H_iv).$$

This is enough to estimate commutator norms, dominant interference planes, and route mismatch.

9.2 Stochastic traces and fingerprints

For a random probe z with $\mathbb{E}[zz^*] = I$,

$$\text{Tr}(A) = \mathbb{E}[z^*Az].$$

Hutchinson estimation therefore gives trace words, curriculum characters, commutator energies, and selected moments using a modest number of operator rollouts [1], [2], [13]. The memory index can store noisy confidence intervals rather than exact invariants.

9.3 Low-rank and structured metrics

Practical metrics may use diagonal, block-diagonal, Kronecker-factored, low-rank-plus-diagonal, or module-local approximations. K-FAC is one established way to obtain tractable curvature blocks in neural networks [14]. In control, balanced truncation or empirical controllability Gramians play a similar role. In probabilistic models, sufficient-statistic Fisher blocks may be available analytically.

The protocol does not require the metric to be exact. It requires the approximation error to be exposed, because a poor metric can create false mechanism matches or destroy path consistency.

9.4 Coarse-to-fine retrieval

A scalable system can use three stages.

1. **Semantic or structural shortlist:** retrieve by content, topology, tags, or context.
2. **Low-degree geometric filter:** compare trace, spectrum, overlap, and commutator sketches.
3. **Local alignment test:** optimize a low-rank intertwiner only for the few surviving candidates.

This avoids comparing every context with every memory item.

9.5 Learning the transport

Several approximations are possible:

- orthogonal Procrustes alignment between dominant information subspaces;
- canonical-correlation or principal-angle matching;
- learned partial isometries regularized by operator intertwining;
- transport predicted by a context encoder and corrected by local probes;
- graph or bundle connections learned jointly with module boundaries;
- sparse routing matrices constrained to preserve a local metric.

Transport confidence should decline near degenerate or highly symmetric operator tuples, where several alignments are equally valid. Repeated identical modules are especially problematic because the memory cannot reliably determine which copy owns a write. Deliberate small heterogeneity or **detuning** of modules can break this symmetry and improve identification and portability [1].

9.6 Memory lifecycle and garbage collection

CMCP requires explicit operations beyond gradient descent:

- **promote:** provisional evidence becomes shared;
- **merge:** two records have equivalent geometry and nonconflicting provenance;
- **split:** one record has persistent multimodal geometry or contradictory intervention signatures;
- **branch:** a mechanism changes over time;
- **freeze:** information and intervention signatures stabilize;
- **reopen:** new conditional information exceeds a threshold;
- **retire:** a record is unused or superseded;
- **quarantine:** provenance, responsibility, or novelty is suspect.

These operations can be implemented through model selection, nonparametric allocation, expert routing, program synthesis, or database transactions. The protocol specifies the evidential tests, not one data structure.

10. A Rank-One Diagnostic Example

Consider two rank-one information operators

$$H_s = a u u^*, \quad H_t = b v v^*, \quad \rho = u^* v,$$

with unit vectors u, v . Two second-order statistics are

$$O_{st} = \frac{\text{Tr}(H_s H_t)}{\|H_s\|_F \|H_t\|_F} = \rho^2,$$

and

$$\frac{\|[H_s, H_t]\|_2}{ab} = |\rho| \sqrt{1 - \rho^2}.$$

They distinguish three geometric cases.

1. **Parallel:** $|\rho| = 1$. The contexts use the same direction and commute. This may be a duplicate manifestation or a new independent observation. Curvature cannot decide; the conditional-information ledger must.
2. **Orthogonal:** $\rho = 0$. The contexts affect independent supports and commute. With correct routing, their updates should be modular.
3. **Oblique:** $0 < |\rho| < 1$. The contexts share a component but encode it differently. The commutator is nonzero and peaks at intermediate overlap. Transport or modular refactorization is required.

The example makes a central point: **curvature is not a universal synonym for redundancy**. Exact duplicate evidence in one fixed direction can be overcounted with zero commutator. Independent orthogonal mechanisms also have zero commutator. The protocol needs the full combination of symmetric overlap, conditional information, and antisymmetric path dependence [4].

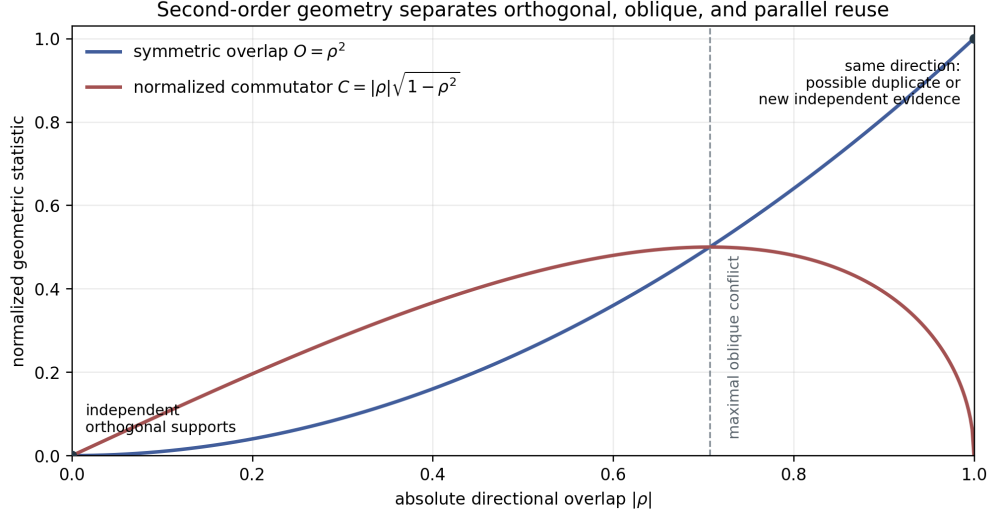


Figure 3: Rank-one symmetric overlap and normalized commutator as a function of directional overlap. Curvature vanishes for both parallel and orthogonal cases, so it must be combined with symmetric overlap and conditional evidence accounting.

11. Diagnostics and Audit

CMCP should be evaluated as a protocol, not only through end-task performance. The following measurements correspond directly to its contracts.

11.1 Evidence-ledger diagnostics

- **Write amplification:** number of shared-memory commits per ground-truth causal innovation.
- **Duplicate idempotence error:** change in memory after replaying an exact or deterministically derived manifestation.
- **Independent-evidence utilization:** information or performance gain from conditionally independent repeated observations.
- **Conditional-information ratio:** $\text{Tr}(F_{c|M}) / (\text{Tr} F_{cc} + \epsilon)$.
- **Overcount and undercount:** deviation of effective precision from a joint or batch reference ledger.
- **Provenance coverage:** fraction of protected writes with reconstructible source, responsibility, and novelty records.

11.2 Routing diagnostics

- precision and recall of selected modules against known intervention structure;
- causal effect of routed versus blocked updates;
- bandwidth and number of active highway routes;
- fraction of credit classified as shared, local, nuisance, new, or drift;
- delayed harm from false-positive shared writes.

11.3 Geometric diagnostics

- operator-tuple match error and transport condition number;
- path-consistency loss across alternative routes;

- pairwise commutator norms;
- interference Gram rank and off-diagonal structure;
- curriculum holonomy angle spectrum;
- symmetric convergence spectra and per-module precision growth.

Interference diagnostics have an important blind spot. A successfully modular system has commuting task operators and therefore little holonomy data. A near-zero shadow cannot, by itself, validate the architecture; it is also consistent with a broken probe or inactive learning. Convergence, information gain, intervention response, and per-module curvature must be measured alongside the shadow [2].

11.4 System-level outcomes

- retention after sequential learning;
- forward transfer and negative transfer;
- adaptation speed under genuine mechanism drift;
- calibration of uncertainty or confidence;
- robustness to adversarial duplicate records and memory poisoning;
- capacity growth, module reuse, and garbage-collection quality;
- curriculum-order sensitivity outside designated shared modules.

12. Broad Application Patterns

CMCP is intended as a protocol layer that can sit above many learning rules.

Domain	Persistent causal memory	Credit token	Information geometry	Typical local state
Continual supervised learning	Shared feature, expert, or mechanism modules	Loss gradient, prediction residual, constraint signal	Fisher, GGN, task Hessian	Adapter, task head, episodic buffer
Predictive coding and energy models	Stable generative transitions	Local prediction-error or distributional residual	Energy Hessian, Fisher, error-coordinate metric	Latent state, local transition parameters
Reinforcement learning	Dynamics factors, skills, options, reward mechanisms	TD innovation, advantage, model-prediction error	Policy Fisher, successor features, transition Jacobian Gram	Episode memory, context policy, eligibility trace
World models and robotics	Object dynamics, contact laws, sensor models	Counterfactual prediction error, control residual	Controllability/observability Gramians, Gauss-Newton	Task controller, regime adapter
Federated and multi-domain learning	Cross-site invariant mechanisms	Site-local gradient or sufficient-statistic increment	Site Fisher/operator tuple	Private site adapter and provenance ledger
Scientific model revision	Reusable laws, submodels, latent mechanisms	Posterior score, residual, failed prediction	Fisher information, sensitivity Hessian	Dataset-specific nuisance model or hypothesis branch

Domain	Persistent causal memory	Credit token	Information geometry	Typical local state
Autonomous agents	Skills, tools, rules, semantic and episodic memory	Critique, failed plan, tool error, reward signal	Jacobian/Hessian of outcome sensitivity	Conversation, episode, user, or tool-local memory
Distributed databases and knowledge systems	Canonical causal event or entity records	New record, contradiction, update proposal	Conditional covariance and dependency operator	Source-specific record, quarantine, version branch

12.1 Continual learning

Regularization methods such as elastic weight consolidation and synaptic intelligence protect directions judged important to prior tasks [15], [16]. Gradient projection methods constrain new updates against episodic memories [17]. Progressive and modular systems isolate parameters or add new capacity [18], [19]. CMCP is complementary. It asks whether several tasks concern the same underlying mechanism, whether their evidence is new, how their coordinates should be aligned, and whether shared or local storage is warranted.

12.2 Reinforcement learning and world models

A scalar TD error mixes many causal sources: transition surprise, reward-model error, policy mismatch, state aliasing, and stochasticity [20], [21]. CMCP would factorize the innovation, route model error to the responsible dynamics component, route policy error to the responsible skill or option, and use conditional information to avoid repeatedly consolidating the same trajectory. Persistent memory can store environment mechanisms, while local state stores episode- or regime-specific adaptations.

12.3 Control and robotics

Controllers often reuse physical mechanisms across tasks while encountering changing payloads, contacts, sensors, or environments. A local sensitivity or controllability geometry can match mechanisms across robot configurations. A credit-only highway can update the responsible dynamics or calibration module without adding a permanent shortcut to the control law. Drift branching is especially important: a changed actuator should create a versioned mechanism rather than silently corrupt the nominal model.

12.4 Federated and multi-domain systems

Federated learning needs to distinguish globally shared structure from site-specific effects. Raw gradient aggregation can overcount correlated populations or erase minority regimes. CMCP can treat each site update as an evidence packet, use operator fingerprints to identify shared mechanisms, residualize against the global ledger, and keep local residuals private. Trace and operator sketches may be easier to communicate than raw data, although privacy leakage from second-order statistics requires separate protection.

12.5 Scientific model updating

Scientific evidence often enters through pipelines that derive many observables from the same experiment. Treating each derived record as independent can create false confidence. CMCP’s conditional-information

ledger offers a local way to discount predictable descendants while retaining independent replication. Geometric fingerprints can identify when different experimental regimes constrain the same mechanism in different coordinates. Intervention responsibility corresponds to experimental manipulation or credible natural experiments.

12.6 Autonomous agents and long-term memory

Memory-augmented agents frequently use semantic similarity for retrieval and heuristics for consolidation. CMCP adds a distinction between semantic resemblance and evidential novelty. A retrieved rule may be useful in many episodes without being rewritten after each use. A failed action can generate credit to the responsible skill, tool model, or user-specific adapter. Promotion to global memory requires recurrence across contexts and intervention-consistent benefit.

13. Experimental Program

A credible evaluation should begin in synthetic environments with known causal structure, then move to realistic domains.

13.1 Synthetic causal benchmark

Construct a structural causal system with known mechanisms and parameter supports. Generate contexts that separately instantiate:

1. an exact duplicate manifestation of one innovation;
2. a deterministic descendant of stored evidence;
3. an independent repeated measurement of the same mechanism;
4. the same mechanism expressed through a rotated representation;
5. independent mechanisms on orthogonal supports;
6. a context-local exception;
7. a genuinely new mechanism;
8. gradual mechanism drift;
9. abrupt regime change;
10. nuisance correlation with no intervention effect;
11. adversarial attempts to trigger shared writes.

The benchmark should provide ground truth for mechanism identity, causal responsibility, evidence dependence, and drift.

13.2 Factorial ablations

Compare at least:

1. local learning without shared memory or highways;
2. shared memory with ordinary global gradients;
3. a raw full-error highway without a ledger;
4. a causally routed highway without persistent memory;
5. persistent memory with semantic matching but no information geometry;
6. CMCP without conditional information, using only subspace projection;
7. CMCP without intervention-sensitive responsibility;

8. full CMCP.

These ablations isolate the contribution of reach, memory, geometry, causal routing, and evidence accounting.

13.3 Core predictions

- A full-error highway improves adaptation speed but broadens cross-task interference.
- Memory without a highway reduces duplicate writes but leaves difficult long-range credit problems.
- A causal highway without persistent memory lacks a stable reference for novelty and repeatedly rediscovers what is already known.
- Subspace projection alone suppresses legitimate independent repeated evidence.
- Conditional-information residualization preserves independent evidence while suppressing deterministic duplicates and correlated descendants.
- Operator alignment reduces false mechanism splits under rotated representations.
- Full CMCP yields low write amplification, high router precision, localized interference, better retention, and faster adaptation to true novelty.
- Under successful modularization, holonomy diagnostics approach zero while symmetric information gain and intervention response remain nonzero.

13.4 Stress tests

The protocol should be stressed under:

- poor curvature approximations;
- highly symmetric interchangeable modules;
- nonstationary metrics;
- delayed or sparse intervention feedback;
- conflicting provenance;
- capacity limits;
- long loops far from a local quadratic regime;
- partially observed causes;
- strategic agents that manipulate the write gate.

A negative result should identify which contract failed. For example, high write amplification with low path error points to a conditional-information failure, while order sensitivity with correct multiplicity points to a transport or modularity failure.

14. Relationship to Existing Approaches

CMCP is not a replacement for standard learning algorithms. It is a protocol for deciding where and how their updates become persistent knowledge.

14.1 Natural gradient and curvature-aware optimization

Natural gradient makes a local update invariant to smooth reparameterization and corrects metric anisotropy [9]. K-FAC and related approximations make this practical [14]. CMCP uses the same geometry for mechanism matching, transport, and evidence units. Natural gradient answers *how to measure and scale a local*

update. CMCP additionally asks *whether the update is new, which memory owns it, and whether it should be shared*.

14.2 Regularization, replay, and projection

EWC, synaptic intelligence, replay, GEM, and related methods preserve prior performance by penalizing or projecting updates [15], [16], [17], [22]. They mainly control displacement in important directions. CMCP adds a causal evidence ledger and a mechanism identity layer. It can use replay or projection as a local realization of a write decision, but it seeks to prevent redundant evidence from entering the protected state in the first place.

14.3 Modular networks and mixture-of-experts

Mixture-of-experts and progressive architectures allocate specialized capacity and sparse routing [18], [23], [24]. CMCP supplies criteria for deciding when two experts should share a mechanism, when a residual should remain local, when modules should merge or split, and when replicated experts are too symmetric to identify reliably.

14.4 Causal representation learning

Causal representation learning seeks variables and modules aligned with autonomous mechanisms [25], [26]. CMCP assumes that such alignment may be partial and uses interventions plus geometric matching to improve it online. It is closer to a protocol for maintaining and updating causal representations than to a theorem that discovers them from observational data alone.

14.5 Predictive coding, local learning, and feedback channels

Predictive coding and local energy-based learning provide biologically and computationally attractive local discrepancy signals [27], [28]. Their long-range credit can be limited by local self-adjoint inference geometry [3]. Direct feedback and error highways provide reach [8]. CMCP combines direct credit reach with causal factorization, novelty filtering, and metric-aware transport so that the solution to credit starvation does not become diffuse entanglement.

14.6 Bayesian filtering and sufficient-statistic accumulation

Bayesian filtering and Kalman-style updates already separate prior state from new evidence [29]. CMCP generalizes the idea to modular, context-dependent systems where the main challenge is deciding whether an observation addresses the same mechanism, how to transport it into a common frame, and whether it is conditionally new. In exponential-family cases, the memory ledger can be implemented directly through natural parameters and sufficient statistics. In non-Bayesian systems, the same structure can govern gradients, constraints, or program revisions.

15. Limitations and Open Problems

15.1 Second-order geometry is local

Two mechanisms can agree to second order at an operating point and differ at higher order. Strongly nonlinear or far-from-equilibrium learning may require third-derivative tensors, higher jets, nonlinear transport, or

direct comparison of finite update maps. A natural extension is a hierarchy of operator-and-tensor signatures rather than a single Hessian tuple [1].

15.2 Geometry does not establish causal direction by itself

Operator overlap and curvature identify shared sensitivity structure. They do not prove that a module is an upstream cause rather than a correlated proxy. Intervention, randomized perturbation, environment variation, or a trusted structural model remains necessary. In domains where interventions are impossible, causal responsibility will be uncertain and writes should remain provisional.

15.3 Conditional Fisher information is an approximation

The Schur-complement rule captures linear conditional predictability of score statistics. Nonlinear redundancy can remain. Exact conditional mutual information is more general but harder to estimate. Memory summaries can also lose dependence information, causing false novelty. The protocol should report the resolution and confidence of its ledger.

15.4 Symmetry and identifiability

Interchangeable modules produce multiple equally valid transports. This can cause write races, unstable ownership, and poor portability certificates. Detuning, explicit addresses, provenance, or a symmetry-aware quotient representation may be necessary.

15.5 Memory ossification and concept drift

A stable read-many/write-gated core can become too difficult to revise. Thresholds should depend on evidence volume, mechanism age, drift indicators, and the cost of false consolidation. Versioning and reversible provisional writes are essential.

15.6 Router self-confirmation

A module that already receives credit may appear increasingly responsible because it changes the output. Controlled randomized probes and held-out intervention tests are needed to prevent this feedback loop.

15.7 Computational and privacy cost

Second-order sketches, intervention probes, and provenance logs consume compute and storage. In federated or privacy-sensitive settings, Fisher and gradient sketches may leak information. Secure aggregation, differential privacy, or local audit mechanisms must be integrated separately.

15.8 The protocol may discover useful mechanisms that are not ontologically causal

A module can be intervention-useful and stable without corresponding to a unique real-world causal variable. CMCP's operational criterion is enough for credit and memory management, but stronger scientific interpretation requires external validation.

16. Conclusion

The Causal Memory and Credit Protocol is a proposal for turning adaptive learning into disciplined evidence assimilation. Its central architectural idea is simple:

Represent a shared mechanism once, let many contexts read it, allow only causally responsible and conditionally novel evidence to modify it, and transport that evidence through a sparse credit channel that respects local information geometry.

Persistent memory and a causal credit-only highway are complementary. The memory gives the highway a stable answer to *what is already known*. The highway gives the memory a selective answer to *which distant mechanism should be allowed to change*. Information geometry supplies the language for recognizing the same mechanism across representations, learning transport maps, and auditing path dependence. Conditional information supplies the missing distinction between redundant manifestations and independent repeated evidence.

The protocol therefore separates four operations that ordinary learning often conflates:

1. computational reuse of a mechanism;
2. causal assignment of a discrepancy;
3. geometric transport of credit;
4. statistical assimilation of innovation.

This separation is broader than any particular neural or Bayesian implementation. It applies wherever a system learns sequentially, shares mechanisms across contexts, and risks either starving distant modules of credit or rewriting shared state with diffuse and redundant evidence. The decisive empirical question is not whether CMCP can produce another accuracy gain. It is whether it can achieve the protocol contracts: idempotence for duplicate manifestations, accumulation for independent evidence, commutation for independent mechanisms, coordinate-equivariant matching, bounded path inconsistency, sparse responsible writes, and robust adaptation to genuine novelty.

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